



US 20250286213A1

(19) **United States**(12) **Patent Application Publication**
PARK(10) **Pub. No.: US 2025/0286213 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **BATTERY PACK AND DEVICE INCLUDING THE SAME****Publication Classification**(51) **Int. Cl.***H01M 50/383* (2021.01)*H01M 50/204* (2021.01)*H01M 50/358* (2021.01)*H01M 50/673* (2021.01)(52) **U.S. Cl.**CPC *H01M 50/383* (2021.01); *H01M 50/204*(2021.01); *H01M 50/358* (2021.01); *H01M**50/673* (2021.01)(71) Applicant: **LG ENERGY SOLUTION, LTD.**,
Seoul (KR)(72) Inventor: **Jong Bum PARK**, Daejeon (KR)(21) Appl. No.: **18/875,707**(22) PCT Filed: **Jul. 13, 2023**(86) PCT No.: **PCT/KR2023/095029**

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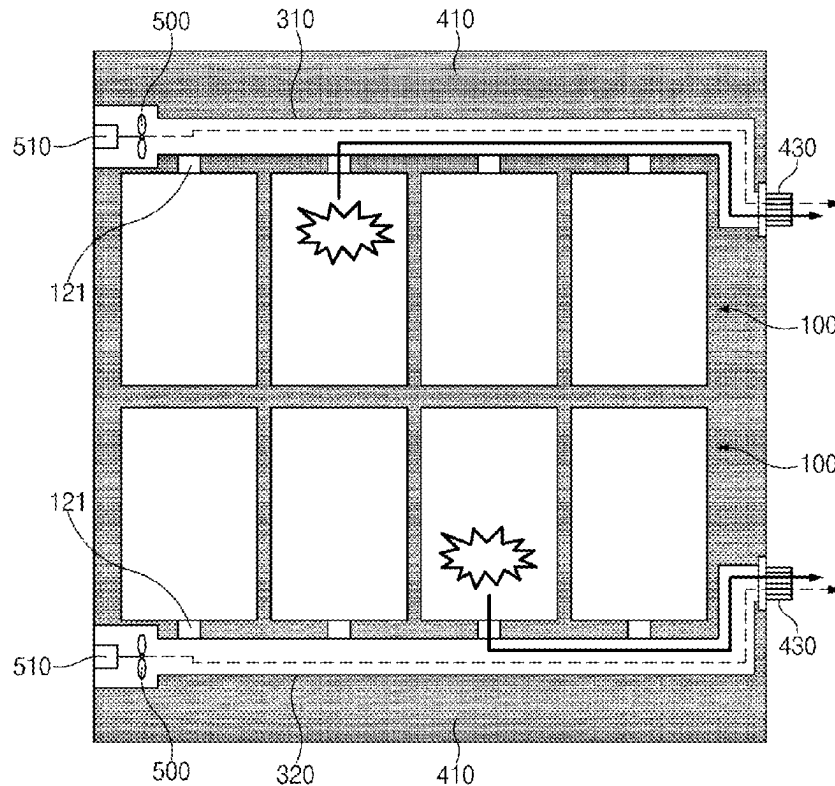
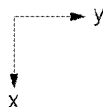
(2) Date: **Dec. 16, 2024**(30) **Foreign Application Priority Data**

Jul. 14, 2022 (KR) 10-2022-0087178

(57)

ABSTRACT

The present disclosure relates a battery pack and a device including the same. The battery pack may include: a plurality of battery modules; a passage frame disposed along an edge of at least a portion of the plurality of battery modules to define an inner passage; and a blowing member communicating with one end of the inner passage of the passage frame to control a gas discharge rate within the inner passage.

300 { 310
320

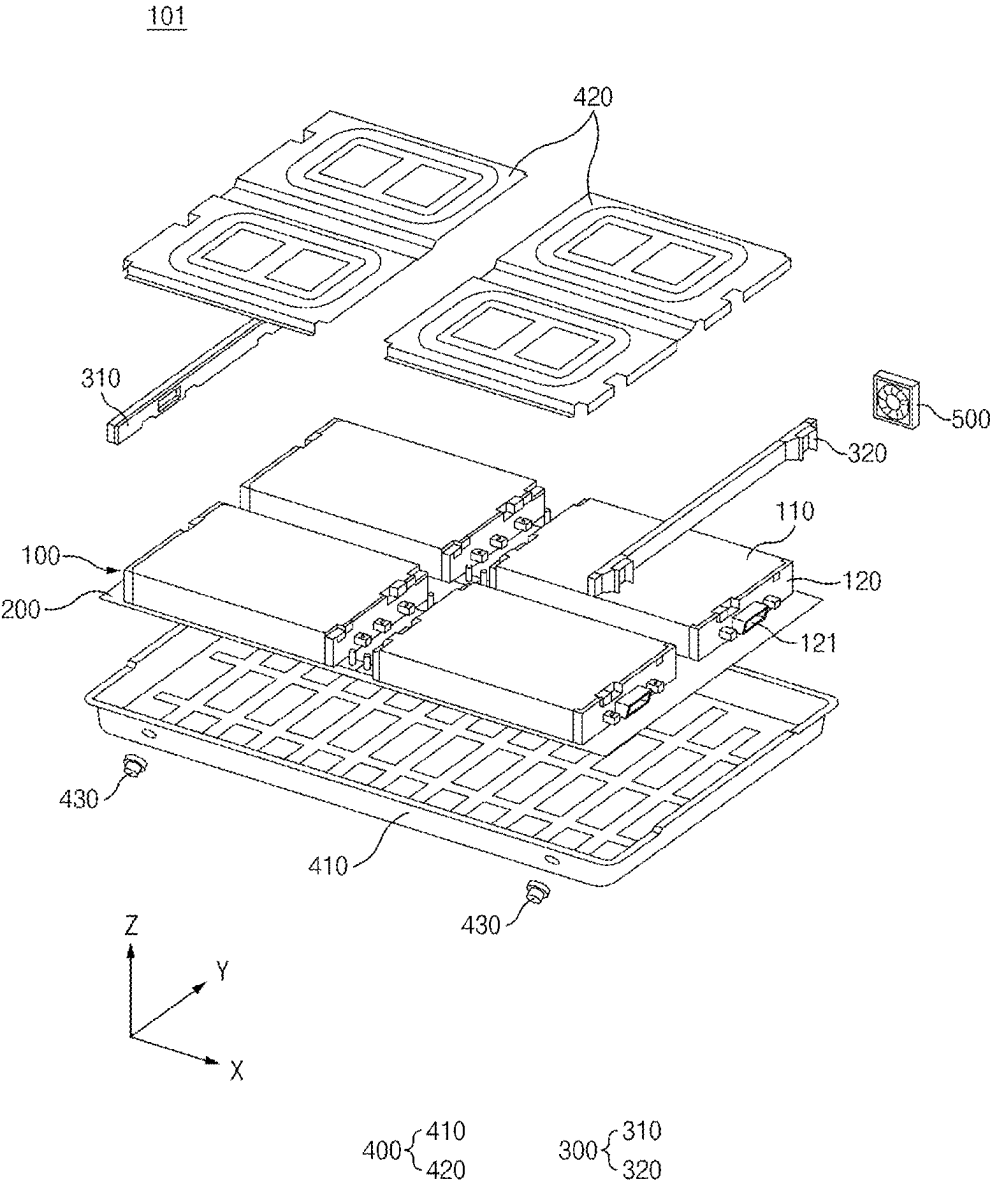


FIG.1

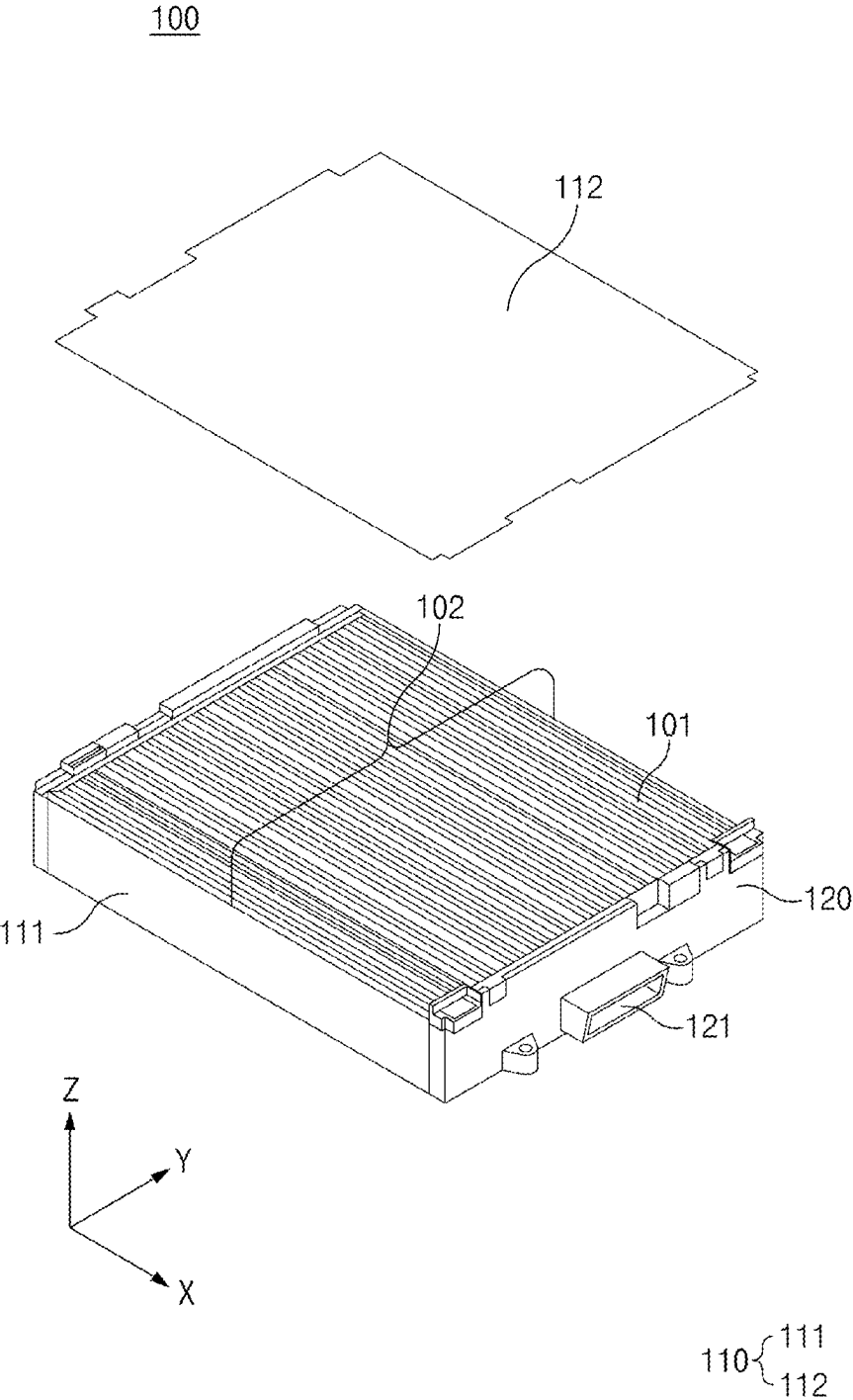


FIG.2

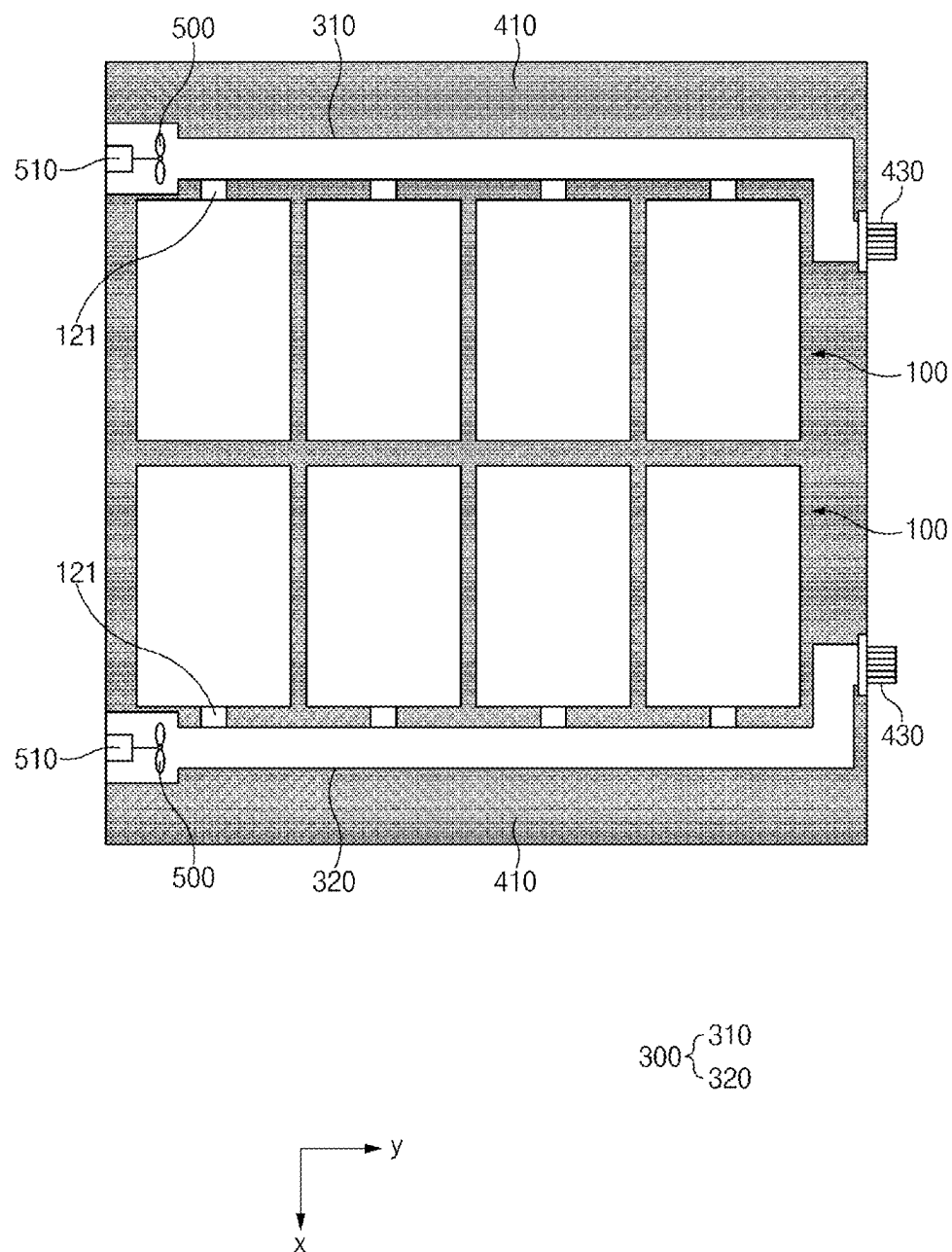


FIG. 3

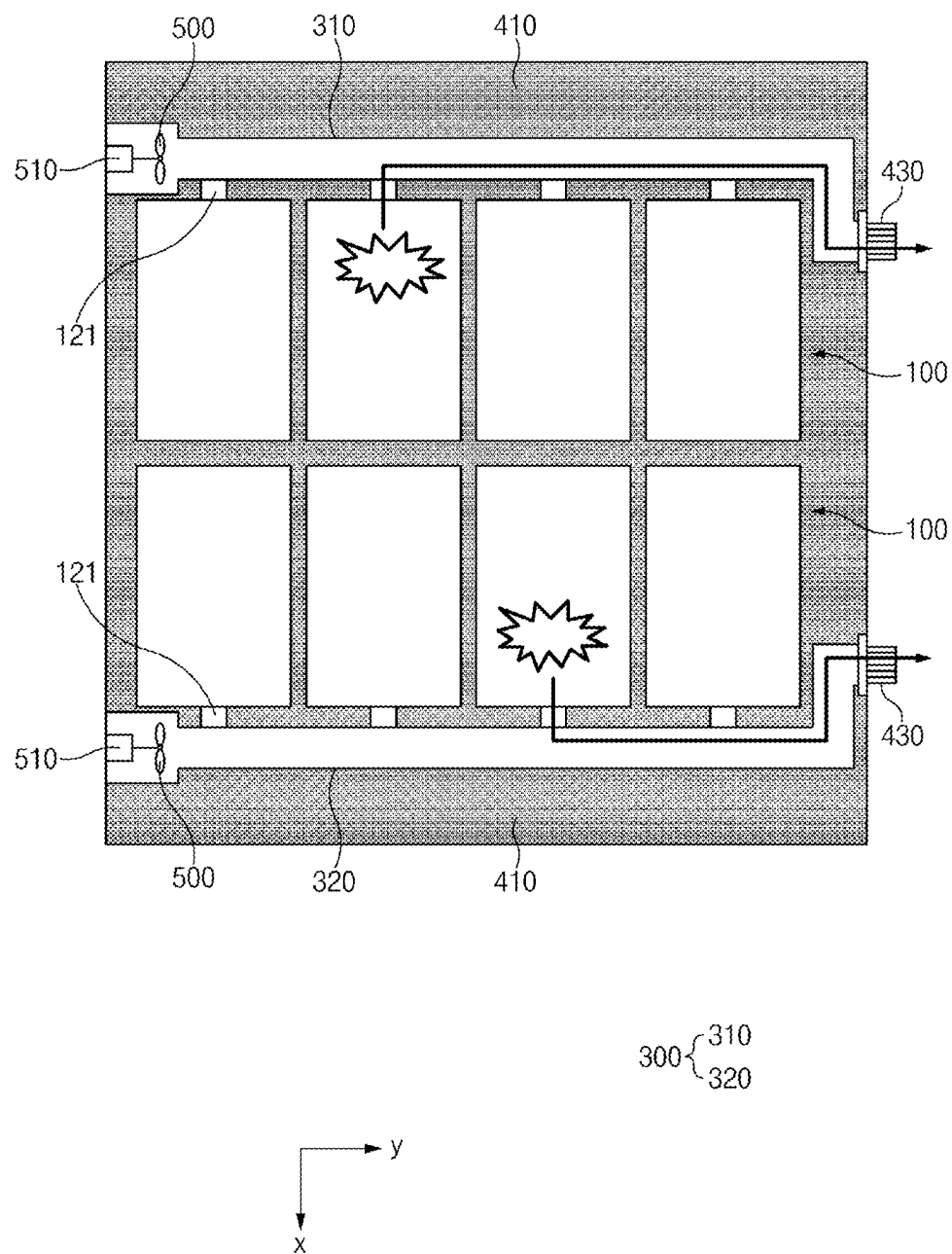


FIG. 4A

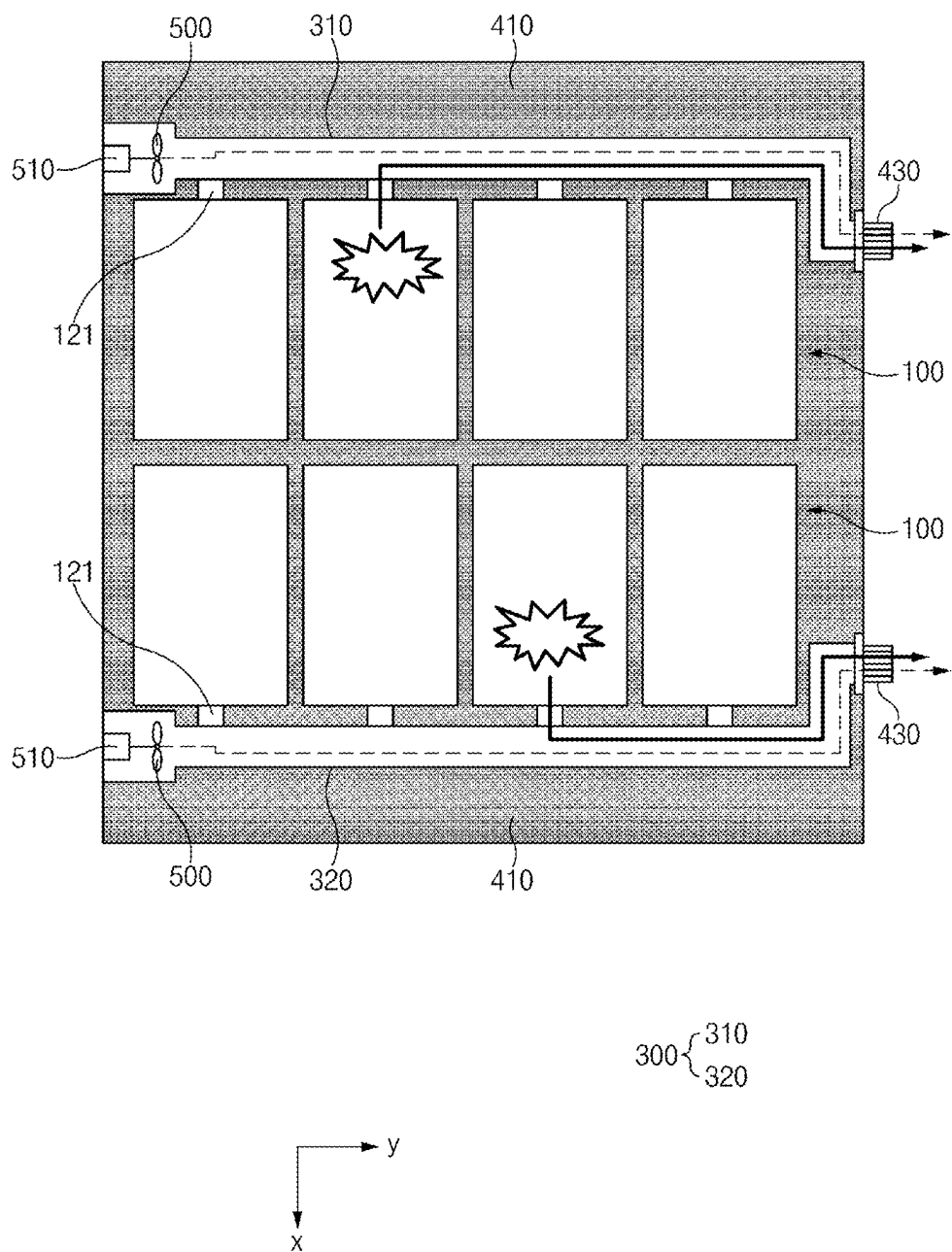


FIG. 4B

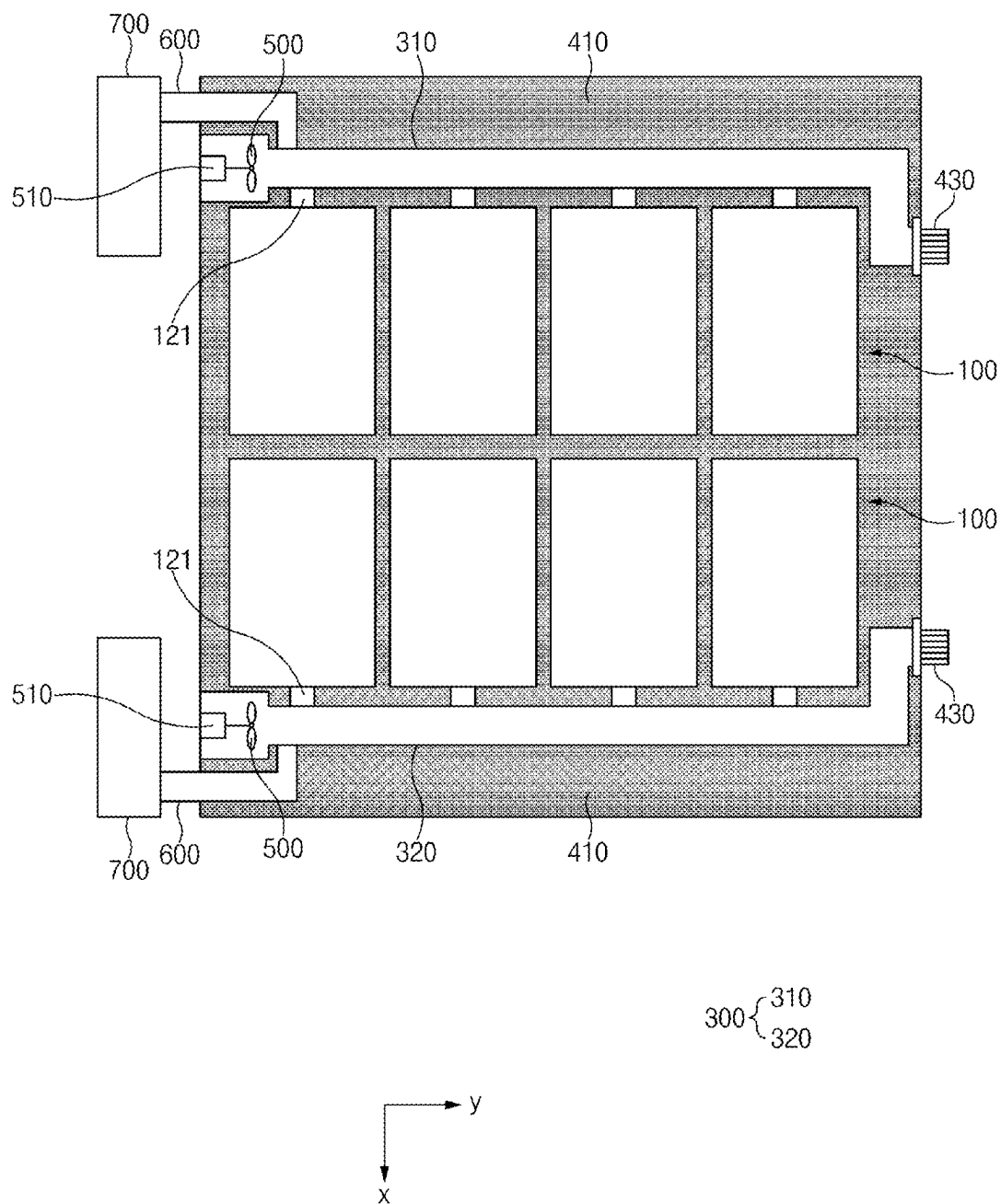


FIG. 5

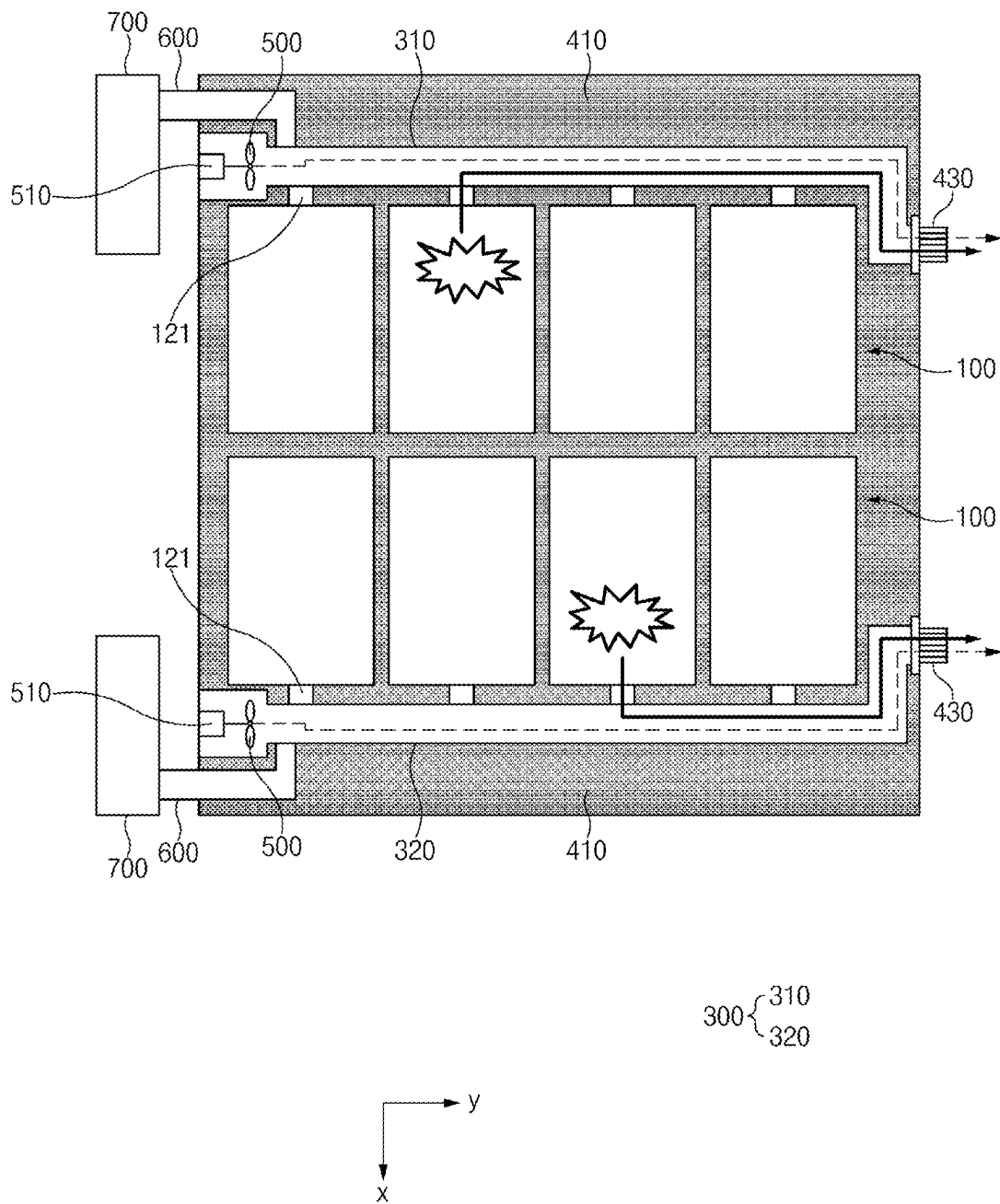
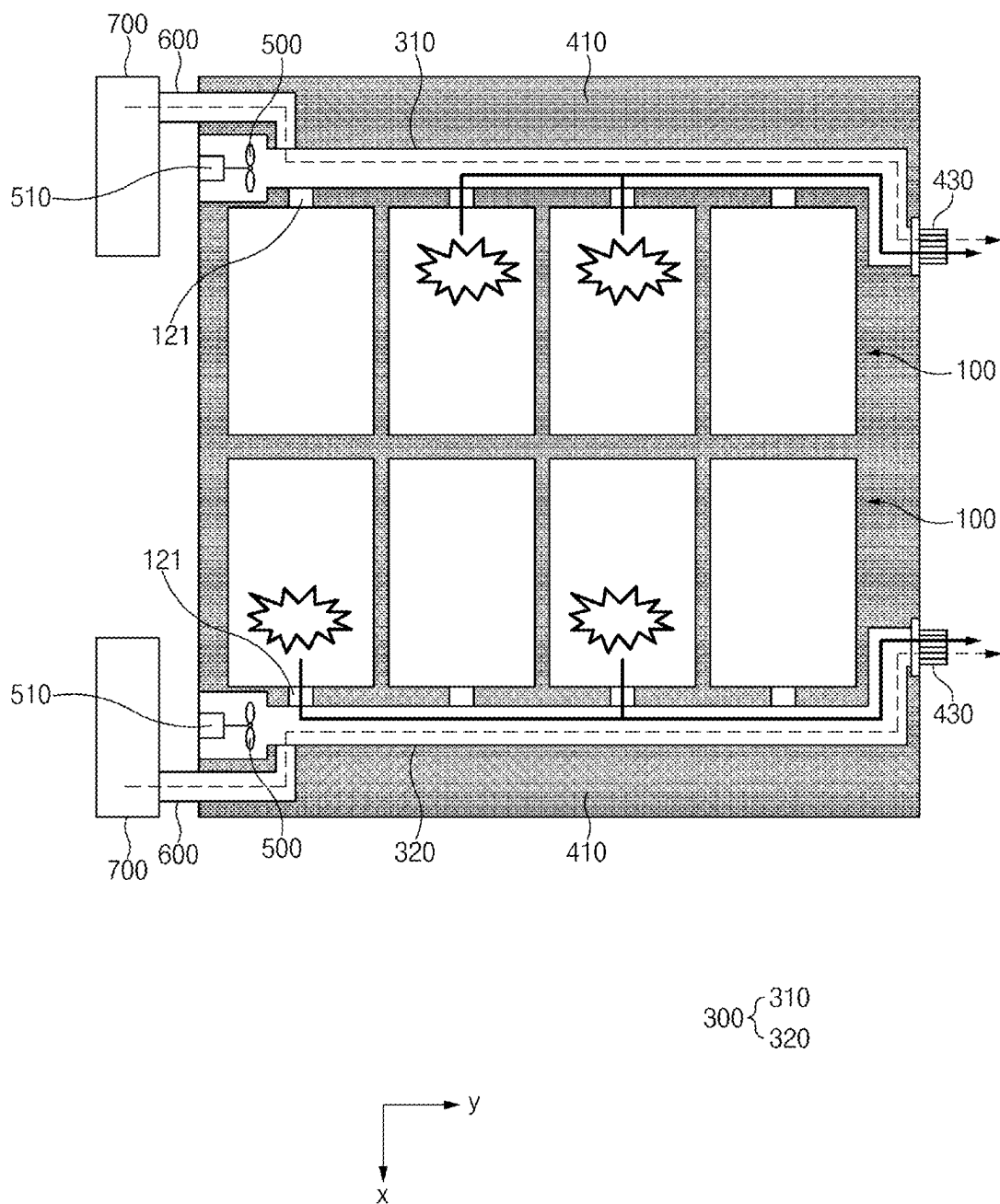


FIG. 6A



BATTERY PACK AND DEVICE INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is a National Phase entry pursuant to 35 U.S.C. § 371 of International Application No. PCT/KR2023/095029, filed on Jul. 13, 2023, and claims the benefit of and priority to Korean Patent Application No. 10-2022-0087178, filed on Jul. 14, 2022, the disclosures of which are incorporated by reference in their entirety for all purposes as if fully set forth herein.

TECHNICAL FIELD

[0002] The present disclosure relates to a battery pack and a device including the same, and more particularly, to a battery pack capable of smoothly discharging a high-temperature gas and a device including the same.

BACKGROUND

[0003] Recently, as demands for portable electronic products such as laptops, video cameras, and mobile phones have rapidly increased, and development of electric vehicles, energy storage batteries, robots, satellites, etc. is regularized, a lot of researches have been conducted on a secondary battery used as a driving power source.

[0004] As needs for structures of large-capacity secondary batteries, demands for battery packs having a multi-module structure in which battery modules, in which a plurality of secondary batteries are connected to each other, are assembled are increasing.

[0005] When a battery cell provided in the battery pack is ignited, a high-temperature gas and flame may be generated. If the high-temperature gas and flame are not discharged smoothly, a pressure inside the battery pack may increase, and thus, the pack structure may collapse.

[0006] The background description provided herein is for the purpose of generally presenting context of the disclosure. Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in this section.

SUMMARY

Technical Problem

[0007] An embodiment of the present disclosure is to provide a battery pack capable of smoothly discharging a gas generated therein and a device including the same.

[0008] The object of the present disclosure is not limited to the aforesaid, but other objects not described herein will be clearly understood by those skilled in the art from descriptions below.

Technical Solution

[0009] A battery pack according to an embodiment of the present disclosure may include: a plurality of battery modules; a passage frame disposed along an edge of at least a portion of the plurality of battery modules to define an inner passage; and a blowing member communicating with one end of the inner passage of the passage frame to control a gas discharge rate within the inner passage.

[0010] According to an embodiment, the battery pack may further include an outlet communicating with the other end of the inner passage.

[0011] According to an embodiment, the battery pack may further include: a pack housing accommodating the plurality of battery modules and the passage frame; a storage tank disposed at at least one of inside or outside of the pack housing; and a fluid supply tube disposed between the storage tank and the passage frame, wherein a fluid stored in the storage tank may be supplied into the inner passage of the passage frame through the fluid supply tube.

[0012] According to an embodiment, the fluid may be at least one of a liquid comprising water or a gas comprising nitrogen.

[0013] According to an embodiment, the blowing member may operate when a pressure of the inner passage reaches a first pressure, and the storage tank may operate when the pressure of the inner passage reaches the first pressure or a second pressure higher than the first pressure after the blowing member operates.

[0014] According to an embodiment, the storage tank may discharge the fluid at a third pressure higher than the second pressure.

[0015] According to an embodiment, the fluid supply tube may be branched from the passage frame.

[0016] According to an embodiment, the storage tank may be provided in plurality, one of the plurality of storage tanks may store a gas for discharging foreign substances, which is injected into the inner passage, and the other of the plurality of storage tanks may store a liquid for discharging foreign substances, which is injected into the inner passage.

[0017] According to an embodiment, the passage frame may include: a first passage frame disposed along an edge of one side of the plurality of battery modules; and a second passage frame disposed along an edge of the other side of the plurality of battery modules.

[0018] According to an embodiment, the blowing member may include a rotating fan.

[0019] A device according to an embodiment of the present disclosure may include the above-described battery pack.

Advantageous Effects

[0020] According to the embodiment of the present disclosure, the gas generated in the battery pack may be smoothly discharged through the blowing member disposed inside the battery pack to suppress the generation of the flame and the increase in pressure inside the battery pack.

[0021] According to the embodiment of the present disclosure, the foreign substances accumulated inside the battery pack may be discharged to the outside through the fluid injected into the battery pack to manage the pressure inside the battery pack to the certain level.

[0022] According to the embodiment of the present disclosure, when the abnormality of the battery cell occurs, the stability of the battery pack may be improved through at least one of the blowing member disposed inside the battery pack or the fluid injected into the battery pack.

[0023] In addition, the various effects that are directly or indirectly identified through the present disclosure may be provided.

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The accompanying drawings illustrate a preferred embodiment of the present disclosure and together with the foregoing disclosure, serve to provide further understanding of the technical features of the present disclosure, and thus the present disclosure is not construed as being limited to the drawing.

[0025] FIG. 1 is a perspective view of a battery pack according to a first embodiment of the present disclosure.

[0026] FIG. 2 is an exploded perspective view of a battery module illustrated in FIG. 1.

[0027] FIG. 3 is a cross-sectional view for specifically explaining a passage frame illustrated in FIG. 1.

[0028] FIGS. 4A and 4B are views for explaining a method for discharging a gas within a battery pack according to an embodiment of the present disclosure, which uses the passage frame illustrated in FIG. 3.

[0029] FIG. 5 is a cross-sectional view illustrating a portion of a battery pack according to a second embodiment of the present disclosure.

[0030] FIGS. 6A and 6B are views for explaining a method for discharging a gas within a battery pack according to another embodiment of the present disclosure, which uses the passage frame illustrated in FIG. 5.

DETAILED DESCRIPTION

[0031] Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings so that those of ordinary skill in the art can easily carry out the present disclosure. However, the present disclosure may be implemented in several different forms and is not limited or restricted by the following embodiments.

[0032] In order to clearly explain the present disclosure, detailed descriptions of portions that are irrelevant to the description or related known technologies that may unnecessarily obscure the gist of the present disclosure have been omitted, and in the present specification, reference symbols are added to components in each drawing. In this case, the same or similar reference numerals are assigned to the same or similar elements throughout the specification.

[0033] Also, terms or words used in this specification and claims should not be restrictively interpreted as ordinary meanings or dictionary-based meanings, but should be interpreted as meanings and concepts conforming to the scope of the present disclosure on the basis of the principle that an inventor can properly define the concept of a term to describe and explain his or her invention in the best ways.

First Embodiment

[0034] FIG. 1 is a perspective view of a battery pack according to a first embodiment of the present disclosure, and FIG. 2 is an exploded perspective view of a battery module illustrated in FIG. 1.

[0035] Referring to FIG. 1, a battery pack 101 according to the first embodiment of the present disclosure may include a pack housing 400, a plurality of battery modules 100 accommodated in the pack housing 400, and a passage frame 300.

[0036] As illustrated in FIG. 2, each of the plurality of battery modules 100 may include a battery cell stack 102 in

which a plurality of battery cells 103 are stacked, a module frame 110 accommodating the battery cell stack 102, and an endplate 120.

[0037] The plurality of battery cells 103 may be stacked to be electrically connected to each other, thereby providing the battery cell stack 102. For example, the plurality of battery cells 103 may be stacked in a direction parallel to a y-axis as illustrated in FIG. 2.

[0038] The module frame 110 that accommodates the battery cell stack 102 may include an upper plate 112 and a lower frame 111. The lower frame 111 may be a U-shaped frame. The U-shaped lower frame 111 may include a bottom part and two side parts respectively extending from both ends of the bottom part in a z-axis direction. The bottom part may cover a bottom surface of the battery cell stack 102 (in a direction opposite to the z-axis), and the side parts may cover both side surfaces of the battery cell stack 102 (in the y-axis direction and a direction opposite to the y-axis direction).

[0039] The upper plate 112 may be provided as a single plate-shaped structure that surrounds a top surface (z-axis direction) except for the bottom surface, which is surrounded by the lower frame 111, and both the side surfaces. The upper cover 112 and the lower frame 111 may provide a structure that is coupled by welding to cover the battery cell stack 102 in a vertical direction and in a left and right direction in a state of being in contact with edge portions corresponding to each other. The battery cell stack 102 may be physically protected through the upper plate 112 and lower frame 111. Each of the upper plate 112 and the lower frame 111 may include a metal material with predetermined strength.

[0040] The endplate 120 may cover the battery cell stack 102 exposed to both ends of the module frame 110. A venting gate 121 may be provided in at least one of the two endplates 120. The venting gate 121 may communicate with the inside of the battery module 100 to release flame or heat generated inside the battery module 100. The venting gate 121 may be connected to an opening (not shown) defined in a portion of the endplate 120 to communicate with the inside of the battery module 100. The venting gate 121 may be disposed to face the outside of the battery pack 101.

[0041] The plurality of battery modules 100 may be arranged in a row or in a matrix form. For example, some of the plurality of battery modules 100 arranged in the matrix form may be arranged in a first row parallel to the y-axis direction, and some of the remaining battery modules 100 may be arranged in a second row parallel to the first row. The plurality of battery modules 100 arranged in the first row and the plurality of battery modules 100 arranged in the second row may be arranged symmetrically with respect to the y-axis direction. The venting gate 121 of each of the plurality of battery modules 100 arranged in the first row may be disposed in a direction opposite to the x-axis direction. The venting gate 121 of each of the plurality of battery modules 100 arranged in the second row may be disposed in the x-axis direction.

[0042] The passage frame 300 may be provided along an edge of at least a portion of the plurality of battery modules 100. The passage frame 300 may be provided along at least one side of a plurality of sides of the lower housing 410. The passage frame 300 may be provided in a tubular shape between the lower housing 410 and the battery module 100. At least one passage frame 300 may be provided between the

lower housing 410 and the battery module 100. For example, when the plurality of battery modules 100 arranged in the matrix form are disposed in two rows, the passage frame 300 may include a first passage frame 310 and a second passage frame 320, which are disposed side by side. The first passage frame 310 may be provided to extend along the endplates 120 of the plurality of battery modules 100 arranged in the first row. The second passage frame 320 may be provided to extend along the endplates 120 of the plurality of battery modules 100 arranged in the second row. An inner passage of the first passage frame 310 and an inner passage of the second passage frame 320 may be provided to communicate with each other or may be provided to be spaced apart from each other.

[0043] The pack housing 400 may include a lower housing 410 and an upper cover 420.

[0044] The upper cover 420 may be coupled to the lower housing 410 to cover an upper portion of the battery module 100, thereby protecting an overall length inside the pack housing 400.

[0045] The lower housing 410 may include a bottom surface and a side wall extending in the z-axis direction from the edge of the bottom surface. A pack tray 200 may be seated on the bottom surface. The battery module 100, the passage frame 300, the blowing member 500, and the pack tray 200 may be accommodated in the lower housing 410. The plurality of battery modules 100, the passage frame 300, and the blowing member 500 may be disposed on the pack tray 200. The plurality of battery modules 100 and the passage frame 300 may be fixed to the pack tray 200 as necessary.

[0046] The blowing member 500 may discharge air into the inner passage of the passage frame 300 through an outlet 300. The blowing member 500 may be a blowing fan (or rotating fan) having a plurality of blades installed to rotate about an axis or may be a blowing member without blades.

[0047] The blowing member 500 may operate when at least one of a gas flow rate, a gas amount, and a pressure inside the passage frame 300 reaches a certain level (reference value). Due to the operation (or rotation) of the blowing member 500, the gas in the passage frame 300 may be discharged to the outside at a rate exceeding performance of the venting member to suppress an increase in temperature and pressure inside the battery pack 101. The venting member may include at least one of a venting gate 121, an outlet 430, a venting valve, or a rupture disc.

[0048] The outlet 430 may be disposed at one side wall of the lower housing 410 to discharge heat or flame generated inside. At least one of the venting valve or the rupture disc, which are provided in the venting member, may be disposed around the outlet 430. The rupture disc may be connected to the passage frame 300. The rupture disc may be provided to be ruptured when the pressure of the gas flowing into the inner passage of the passage frame 300 exceeds a certain pressure.

[0049] Various control and protection systems, such as a battery management system (BMS) and a cooling system, together with the battery module 100 may be installed inside the pack housing 400 including the lower housing 410 and the upper cover 420. The battery management system may monitor at least one of a pressure, a temperature, or a gas of at least one of the battery module 100 or the passage frame 300. The battery management system may control whether the blowing member 500 operates by comparing a moni-

tored measurement value to a preset reference value. The battery management system may control the blowing member 500 to operate when the monitored measurement value reaches the preset reference value. The battery management system may control the blowing member 500 so as not to operate when the monitored measurement value is less than the preset reference value.

[0050] As an example, the battery management system may compare the measured pressure to a preset reference pressure and may control the blowing member 500 to operate when the measured pressure reaches the reference pressure. As another example, the battery management system may compare the measured gas amount to a preset reference gas amount and may control the blowing member 500 to operate when the measured gas amount reaches the reference gas amount. As another example, the battery management system may control the blowing member 500 to operate when at least two of the measured values of the pressure, the temperature, and the gas reach preset reference values.

[0051] FIG. 3 is a cross-sectional view for specifically explaining the passage frame illustrated in FIG. 1. There are two passage frames illustrated in FIG. 3, but are not limited thereto, and their positions and number may be appropriately selected as necessary.

[0052] Referring to FIG. 3, the passage frame 300 according to the first embodiment of the present disclosure may communicate with the venting gate 121 and the outlet 430 of the battery module 100. When thermal runaway occurs from the battery module 100, the passage frame 300 may minimize an impact on the surrounding battery modules by inducing the heat and flames to the outside. Here, the flame contained in the generated high-pressure venting gas may be burned while passing through the inner passage of the passage frame 300 and then be discharged to the outside in a safer state. In addition, the passage frame 300 may act as a support frame that stably supports the battery module 100 to improve the stability of the battery pack 101.

[0053] As described above, the passage frame 300 may include a first passage frame 310 and a second passage frame 320. The first passage frame 310 may communicate with the venting gate 121 of each of the plurality of battery modules 100 disposed in the first row among the plurality of battery modules 100. The second passage frame 320 may communicate with the venting gate 121 of each of the plurality of battery modules 100 disposed in the second row among the plurality of battery modules 100.

[0054] The blowing member 500 may be disposed at at least one side of the first passage frame 310 and the second passage frame 320 (e.g., side in the opposite direction of the y-axis). The outlet 430 may be disposed at at least one other side (e.g., side in the direction of the y-axis) of the first passage frame 310 and the second passage frame 320.

[0055] The blowing member 500 may be disposed in the inner passage of each of the passage frames 310 and 320 or may be disposed between each of the passage frames 310 and 320 and the pack housing 410. The blowing member 500 may be disposed adjacent to the battery module 100 that is furthest from the venting member among the plurality of battery modules 100. The blowing member 500 may forcibly discharge air from the inner passage of the passage frame 300 to the outside through the venting member. Since the blowing member 500 rotates with the rotation of the motor 510, a flow of the gas within the passage frames 310 and 320

may be accelerated. As the blowing member 500 rotates, the gas in the passage frames 310 and 320 may move toward the outlet 430 and then may be discharged to the outside of the battery pack 101.

[0056] FIGS. 4A and 4B are views for explaining a method for discharging a gas within a battery pack according to an embodiment of the present disclosure, which uses the passage frame illustrated in FIG. 3.

[0057] As illustrated in FIG. 4A, issues such as overvoltage, overcurrent, or overheating (or heat issues) may occur within at least one battery module 100 among the plurality of battery modules 100. In this case, the high-pressure venting gas may be discharged from the inside of the battery module 100 through the venting gate 121. The high-temperature and high-pressure gas and flame discharged through the venting gate 121 may be introduced into the inner passages of the passage frames 310 and 320. The introduced high-temperature and high-pressure gas and flame may be discharged to the outside through a venting member including at least one of the outlet 300, the rupture disc, or the venting valve. The gas introduced into the inner passages of the flow frames 310 and 320 may be discharged at a first rate.

[0058] Here, the battery management system may monitor at least one of the temperature, the pressure, the amount of gas, or the flow rate of the gas inside the flow frames 310 and 320 in real time or periodically. The battery management system may control the blowing member 500 to operate when at least one of the temperature, the pressure, the amount of gas, or the flow rate of the gas inside the passage frames 310 and 320 reaches a reference value. The blowing member 500 may rotate by power supplied to the motor 510. Due to the rotational movement of the blowing member 500, the high-pressure gas and flame in the inner passage of the passage frame 300 may be forcibly discharged to the outside through the outlet 430, as illustrated in FIG. 4B. The blowing member 500 may discharge high-pressure gas and flame from the inner passage of the passage frame 300 to the outside at a second rate. The second rate may be higher than the first rate (performance of the venting member) at which the high-pressure gas and flame in the inner passage of the passage frame 300 is discharged to the outside through the venting member including the outlet 300. Thus, the increase in temperature and pressure inside the battery pack 101 may be suppressed to prevent structural deformation and collapse of the battery pack 101 from occurring.

Second Embodiment

[0059] FIG. 5 is a cross-sectional view illustrating a portion of a battery pack according to a second embodiment of the present disclosure. The battery pack illustrated in FIG. 5 may have the same components as the battery pack of the first embodiment illustrated in FIGS. 1 and 3, except that it further includes a storage tank 700 and a fluid supply tube 600. Thus, it is obvious that the detailed description of the same components is regarded as the contents of the battery pack of the first embodiment.

[0060] Referring to FIG. 5, the battery pack 501 according to the second embodiment of the present disclosure may include a storage tank 700 and a fluid supply tube 600.

[0061] The fluid supply tube 600 may be provided to be branched from a passage frame 300. The blowing member 500 disposed within the passage frame 300 may not obstruct a flow of fluid injected through the storage tank 700.

[0062] One end of the fluid supply tube 600 may be coupled to the storage tank 700 disposed at at least one of the inside and outside of the pack housing 400. The other end of the fluid supply tube 600 may communicate with the inner passage of the passage frame 300.

[0063] The storage tank 700 may store the fluid capable of being injected into the inner passage of the passage frame 300. The fluid may be periodically filled and stored in the storage tank 700. The storage tank 700 may store at least one of a gas for discharging foreign substances or a liquid for discharging foreign substances. For example, the gas for discharging the foreign substances may contain nitrogen, and the liquid for discharging the foreign substances may contain water.

[0064] The storage tank 700 may include a plurality of storage tanks storing different fluids or the same fluid. For example, the storage tank 700 may include a first storage tank storing a gas containing nitrogen and a second storage tank storing a liquid containing water. The fluid may be supplied to the inside of the fluid supply tube 600 and the passage frame 300 in order or in reverse order of the first storage tank and the second storage tank. A pump (not shown) connected to the storage tank 700 may pump the fluid in the storage tank 700 so that the fluid stored in the storage tank 700 is injected into the fluid supply tube 600.

[0065] The storage tank 700 may inject the fluid at high pressure into the fluid supply tube 600 and the passage frame 300 toward the outlet 430 to remove the foreign substances accumulated inside the passage frame 300 and the outlet 430. The foreign substances may be generated by being carbonized when the battery cell 103 is ignited or may be generated by melting at least one of the venting member including the outlet 430 or the blowing member 500. Thus, the foreign substances may be smoothly discharged to the outside of the battery pack 101 through the fluid injected at the high pressure.

[0066] As an example, when the temperature and pressure inside the passage frame 300 rise in spite of the operation of the blowing member 500 when the blowing member 500 operates, the fluid may be injected into the passage frame 300 through the fluid supply tube 600. In spite of the operation of the blowing member 500, when at least one of the temperature or pressure of the inner passage of the passage frame 300 reaches a preset reference value, the storage tank 700 may inject the fluid into the passage frame 300. Specifically, the amount of gas generated from at least one of the plurality of battery modules 100 may be greater than the amount of gas discharged to the outside due to the operation of the blowing member 500. In this case, the temperature and pressure inside each of the passage frame 300 and the battery module 100 may increase. When flame is generated and propagated due to the increase in temperature and pressure, the foreign substances generated by the carbonization or melting of the structure inside the battery pack may be accumulated in the inner passage and outlet 430 of the passage frame 300. The foreign substances may block the inner passage and the outlet 430 of the passage frame 300 to reduce the amount of gas discharged through the passage frame 300, thereby accelerating the increase in pressure and temperature inside the battery pack 101. Thus, the fluid injected at high pressure by the storage tank 700 may discharge the foreign substances accumulated in the inner passage and outlet 430 of the passage frame 300 to the outside of the battery pack 101.

[0067] As another example, when the temperature and pressure inside the passage frame 300 are below the reference values after the operation of the blowing member 500 is completed, the storage tank 700 may inject the fluid into the passage frame 300 through the fluid supply tube 600. Thus, the foreign substances accumulated inside the passage frame 300 and at the outlet 430 may be smoothly discharged to the outside of the battery pack 101, and thus, the inside of the battery pack 101 may be maintained at a certain pressure level.

[0068] As another example, without the operation of the blowing member 500, after the gas is discharged through the venting gate 121, the passage frame, and the outlet, the storage tank 700 may inject the fluid into the passage frame 300 through the fluid supply tube 600. Thus, the foreign substances accumulated inside the passage frame 300 and at the outlet 430 may be smoothly discharged to the outside of the battery pack 101, and thus, the inside of the battery pack 101 may be maintained at a certain pressure level.

[0069] FIGS. 6Aa and 6B are views for explaining a method for discharging a gas within a battery pack according to another embodiment of the present disclosure, which uses the passage frame illustrated in FIG. 5.

[0070] As illustrated in FIG. 6A, issues such as overvoltage, overcurrent, or overheating (heat issues) may occur within at least one battery module 100 among a plurality of battery modules 100. In this case, a venting gas from the inside of the battery module 100 may be discharged through a venting gate 121. High-temperature and high-pressure venting gas and flame discharged through the venting gate 121 may be introduced into the inside of passage frames 310 and 320. The introduced high-temperature and high-pressure venting gas and flame may be discharged to the outside through a venting member including an outlet 300 at a first rate.

[0071] Here, a battery management system may monitor at least one of a temperature, a pressure, an amount of gas, or a flow rate of a gas inside the flow frames 310 and 320 in real time or periodically. When at least one of the temperature, the pressure, the amount of gas, or the flow rate of gas inside the flow frames 310 and 320 reaches a reference value, the battery management system may control a blowing member 500 to operate. For example, the battery management system may control the blowing member 500 to operate when a pressure in an inner passage of each of the passage frames 310 and 320 increases and reaches a first pressure. The blowing member 500 may rotate by power supplied to a motor 510. Due to the rotational movement of the blowing member 500, the high-pressure gas and flame inside the passage frame 300 may be forcibly discharged to the outside through the outlet 430. The blowing member 500 may discharge high-pressure venting gas and flame inside the passage frame 300 to the outside at a second rate. The second rate may be higher than the first rate (or the performance of the venting member) at which the high-pressure venting gas and flame inside the passage frame 300 are discharged to the outside through the venting member.

[0072] In spite of the operation of the blowing member 500, when at least one of the temperature, the pressure, the amount of gas, or the flow rate of the inner passage of the flow frame 310 and 320 reaches the reference value, as illustrated in FIG. 6B, a storage tank 700 may inject a fluid into the passage frame 300. For example, the storage tank 700 may inject the fluid into the passage frame 300 when the

pressure of the inner passage of the passage frames 310 and 320 is the first pressure or reaches a second pressure higher than the first pressure. The storage tank 700 may inject the fluid at a third pressure that is higher than the second pressure.

[0073] The fluid injected at a high pressure (e.g., third pressure) by the storage tank 700 may discharge foreign substances accumulated in the inner passage and the outlet 430 of the passage frame 300 to the outside of the battery pack 101. Thus, the inside of the battery pack 101 may be maintained at certain temperature and pressure levels.

[0074] The structures of the battery pack 101 described above are not limited to the embodiments described in each drawing, and the structures described in each drawing may be applied in combination. For example, the battery pack 101 may employ only the plurality of blowing members 500 as described in FIGS. 3, 4A, and 4B, employ only the plurality of storage tanks 700 and fluid supply tubes 600 without the blowing member 500 as described in FIGS. 5, 6A, and 6B, or employ a combination of the blowing member 500, the storage tank 700, and the fluid supply tube 600, as described in FIGS. 5, 6A, and 6B.

[0075] In addition, the battery pack 101 described above may be applied to various devices. It may be applied to transportation means such as electric bicycles, electric vehicles, hybrids, etc., but is not limited thereto and may be applied to all various devices capable of using the battery pack 101.

[0076] While the embodiments of the present disclosure have been described with reference to the specific embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the invention as defined in the following claims.

DESCRIPTION OF THE SYMBOLS

- [0077] 101: Battery pack
 - [0078] 100: Battery module
 - [0079] 120: Endplate
 - [0080] 121: Venting gate
 - [0081] 430: Outlet
 - [0082] 500: Blowing member
 - [0083] 510: Motor
 - [0084] 300, 310, 320: Passage frame
 - [0085] 600: Fluid supply tube
 - [0086] 700: Storage tank
1. A battery pack, comprising:
 - a plurality of battery modules;
 - a passage frame disposed along an edge of at least a portion of the plurality of battery modules to define an inner passage; and
 - a blowing member communicating with one end of the inner passage of the passage frame to control a gas discharge rate within the inner passage.
 2. The battery pack of claim 1, further comprising:
 - an outlet communicating with the other end of the inner passage.
 3. The battery pack of claim 1, further comprising:
 - a pack housing accommodating the plurality of battery modules and the passage frame;
 - a storage tank disposed at at least one of inside or outside of the pack housing; and
 - a fluid supply tube disposed between the storage tank and the passage frame,

- wherein a fluid stored in the storage tank is supplied into the inner passage of the passage frame through the fluid supply tube.
4. The battery pack of claim 3, wherein the fluid is at least one of a liquid comprising water or a gas comprising nitrogen.
5. The battery pack of claim 3, wherein the blowing member operates when a pressure of the inner passage reaches a first pressure, and
wherein the storage tank operates when the pressure of the inner passage reaches the first pressure or a second pressure higher than the first pressure after the blowing member operates.
6. The battery pack of claim 5, wherein the storage tank discharges the fluid at a third pressure higher than the second pressure.
7. The battery pack of claim 3, wherein the fluid supply tube is branched from the passage frame.
8. The battery pack of claim 3, wherein the storage tank is provided in plurality,
wherein one of the plurality of storage tanks stores a gas for discharging foreign substances, which is injected into the inner passage, and
wherein the other of the plurality of storage tanks stores a liquid for discharging foreign substances, which is injected into the inner passage.
9. The battery pack of claim 1, wherein the passage frame comprises:
a first passage frame disposed along an edge of one side of the plurality of battery modules; and
a second passage frame disposed along an edge of the other side of the plurality of battery modules.
10. The battery pack of claim 1, wherein the blowing member comprises a rotating fan.
11. A device, comprising:
the battery pack of claim 1.
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